Geometric Infrared Scale of Yang–Mills Theory on a Compact Three-Manifold: Discreteness, Spectral Threshold, and Fibonacci Hierarchy

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We study $SU(N_c)$ Yang–Mills theory on the minimal compact, connected, orientable, boundaryless three-manifold with trivial fundamental group: the three-sphere $S^3(R)$, the unique solution to these topological requirements by the Poincaré–Perelman theorem.

Compactness forces the transverse vector Laplacian to have purely discrete spectrum; the lowest eigenvalue $\lambda_1^V = 3/R^2$ yields a positive geometric spectral threshold

$$\Delta = \frac{\sqrt{3}}{R}.$$

Identifying $R = 1/\Lambda_{\mathrm{QCD}}$ via the confinement condition gives $\Delta = \sqrt{3}\,\Lambda_{\mathrm{QCD}} \approx 520\,\mathrm{MeV}$, consistent with effective gluon masses extracted from Dyson–Schwinger analyses. This result is the *geometric infrared scale* of the linearised theory in finite volume; it is distinct from, and a lower bound on, the full non-perturbative mass gap in infinite volume.

We introduce a Fibonacci coarse-graining of the mode spectrum—the *Fibonacci spectral stratification*—and show it yields a natural minimal-redundancy self-similar partition of the mode labels. Successive band-edge frequencies satisfy $\Omega_{n+1}/\Omega_n \to \varphi$, where $\varphi = (1+\sqrt{5})/2$ is the golden ratio. Applying this hierarchy to two-gluon bound states yields glueball mass ratios converging to φ, with the first non-degenerate ratio $m_3/m_1 = \sqrt{7/3} \approx 1.528$ consistent with quenched lattice QCD to within 10%.

I. INTRODUCTION

A. Setting

This paper studies the infrared spectral properties of $SU(N_c)$ Yang–Mills (YM) gauge theory in finite spatial volume, with the spatial manifold taken to be a compact three-sphere $S^3(R)$. The motivation is spectral-geometric: compactness of the spatial sections forces the transverse vector Laplacian to have a purely discrete spectrum bounded away from zero, generating a positive infrared scale for the gauge field.

The existence of a positive mass gap in four-dimensional $SU(N)$ Yang–Mills theory on $\mathbb{R}^3 \times \mathbb{R}$ is one of the Clay Millennium Prize Problems [1]. Standard QCD predicts confinement; lattice calculations [2, 3] provide numerical evidence for a discrete glueball spectrum; and Dyson–Schwinger analyses [4, 5] extract an effective dynamical gluon mass of $\sim 500\,\mathrm{MeV}$. A geometric mechanism for the infrared scale has not been identified in the literature. The present paper proposes and analyses one such mechanism, valid in finite volume.

B. Topological Setup

Let M be a compact spatial three-manifold on which YM theory is defined. We impose the following minimality conditions, which single out the simplest compact carrier of a discrete gauge-field spectrum:

- **(M1)** M is compact, connected, orientable, and boundaryless.
- **(M2)** $\pi_1(M) = 0$ (trivial fundamental group).
- **(M3)** No additional topological structure is introduced beyond a round Riemannian metric.

By Perelman's proof of the Poincaré conjecture [6–8], the unique closed 3-manifold satisfying (M1)–(M3) is $M \cong S^3$. We work on $S^3(R) \times \mathbb{R}$, where R is the spatial radius.

Remark I.1. The three-torus T^3 has been used as a compact spatial section in the finite-volume YM literature [14]. It fails (M2) ($\pi_1(T^3) = \mathbb{Z}^3$) and has isometry group $SO(3)$ rather than $SO(4) \cong SU(2)_L \times SU(2)_R$. The three-sphere is the more symmetric and topologically minimal alternative.

C. Results

The paper establishes the following.

(1) (*General lemma.*) For any compact, boundaryless Riemannian 3-manifold M, the linearised YM Hamiltonian has purely discrete spectrum with positive infimum $\Delta(M) = \sqrt{\lambda_1^V(M)}$ (Lemma II.1).
(2) (*Geometric infrared scale.*) For $S^3(R)$: $\lambda_1^V = 3/R^2$ and $\Delta = \sqrt{3}/R$. Setting $R = 1/\Lambda_{\mathrm{QCD}}$ gives $\Delta \approx 520\,\mathrm{MeV}$ (Theorem V.1, Corollary V.4).
(3) (*Natural spectral hierarchy.*) The Fibonacci stratification is a natural minimal-redundancy self-similar

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partition of mode labels; successive band-edge frequencies converge to the golden ratio φ (Theorems IV.2–IV.5).

(4) (*Glueball spectrum.*) Fibonacci band-edge states give a glueball mass hierarchy with $m_{n+1}/m_n \to \varphi$, and $m_3/m_1 = \sqrt{7/3} \approx 1.528$ consistent with lattice QCD to within 10% (Theorem VI.1).

Scope. Results (1)–(4) are proved within the *linearised free-mode theory* on $S^3(R) \times \mathbb{R}$. The persistence of the spectral threshold as $R \to \infty$ in the full non-perturbative theory is an open problem (Conjecture V.1).

Conventions. $\hbar = c = 1$. Metric signature $(-,+,+,+)$. $\varphi = (1+\sqrt{5})/2$. Fibonacci: $F_1 = F_2 = 1$, $F_{n+2} = F_{n+1} + F_n$.

II. YANG–MILLS THEORY ON $S^3 \times \mathbb{R}$

A. Action

Let $G = SU(N_c)$ with generators $\mathrm{Tr}[T^a T^b] = \frac{1}{2}\delta^{ab}$ and structure constants f^{abc}. The gauge connection $A_\mu = A^a_\mu T^a$ has field strength

$$F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + g f^{abc} A^b_\mu A^c_\nu. \tag{1}$$

On $S^3(R)\times\mathbb{R}$ with round metric h_{ij} the Yang–Mills action is

$$S = -\frac{1}{2g^2}\int_{\mathbb{R}} dt \int_{S^3} d^3x\, \sqrt{h}\, h^{\mu\nu} h^{\rho\sigma} F^a_{\mu\rho} F^a_{\nu\sigma}. \tag{2}$$

B. General Spectral Lemma

Lemma II.1 (Spectral threshold on a compact 3-manifold)**.** *Let M be a compact, connected, boundaryless Riemannian three-manifold, and let H be the linearised Yang–Mills Hamiltonian on $M \times \mathbb{R}$ in temporal gauge. Then:*

(i) $\mathrm{spec}(H)$ *is purely discrete, bounded from below, and accumulates only at $+\infty$.*

(ii) *The spectral threshold satisfies $\Delta(M) = \sqrt{\lambda_1^V(M)} > 0$, where $\lambda_1^V(M)$ is the lowest positive eigenvalue of the transverse vector Laplacian on M.*

Proof. The transverse vector Laplacian $\triangle_V$ is a self-adjoint elliptic operator of second order on the compact Riemannian manifold M. Compactness of the Sobolev embedding $H^1(M) \hookrightarrow L^2(M)$ (Rellich–Kondrachov theorem, see [11, 12]) implies that the resolvent $(\triangle_V - \mu)^{-1}$ is compact. The spectral theorem for compact self-adjoint operators then gives a purely discrete spectrum $0 < \lambda_1^V \le \lambda_2^V \le \cdots \nearrow \infty$. The Hamiltonian $H = \sum_k \frac{1}{2}(\dot{q}_k^2 + \omega_k^2 q_k^2)$ with $\omega_k = \sqrt{\lambda_k^V}$ has spectrum $\{\sum_k n_k \omega_k : n_k \in \mathbb{Z}_{\ge 0}\}$, with positive infimum $\omega_1 = \sqrt{\lambda_1^V}$. □

Corollary II.2 (Threshold on S^3)**.** *On $S^3(R)$ the lowest transverse vector Laplacian eigenvalue is $\lambda_1^V = 3/R^2$, giving $\Delta(S^3(R)) = \sqrt{3}/R$.*

Proof. By $SU(2)$-representation theory on $S^3 \cong SU(2)$ (Appendix A), transverse vector harmonics at level ℓ transform in the $(\ell/2, \ell/2)$ representation of $SO(4) \cong SU(2)_L \times SU(2)_R$, yielding $\lambda_\ell^V = \ell(\ell+2)/R^2$ for $\ell \ge 1$. Minimum at $\ell = 1$: $\lambda_1^V = 3/R^2$. □

C. Mode Expansion and Hamiltonian

The transverse gauge field is expanded in vector hyperspherical harmonics [9]:

$$A^a_i(t,\mathbf{x}) = \sum_{\ell=1}^{\infty} \sum_{m,m'} q^a_{\ell m m'}(t)\, \mathbf{Y}^{(i)}_{\ell m m'}(\mathbf{x}), \tag{3}$$

with mode frequencies

$$\boxed{\omega_\ell = \frac{\sqrt{\ell(\ell+2)}}{R}, \quad \ell \ge 1,} \tag{4}$$

and linearised Hamiltonian $H = \sum_{\ell,m,m',a} \frac{1}{2}[(\dot{q}^a_{\ell m m'})^2 + \omega_\ell^2 (q^a_{\ell m m'})^2]$.

Theorem II.3 (Discrete spectrum on S^3)**.** *The Yang–Mills Hamiltonian on $S^3(R) \times \mathbb{R}$ has purely discrete spectrum with positive spectral threshold $\Delta = \sqrt{3}/R$, as a specialisation of Lemma II.1 to $M = S^3(R)$.*

Remark II.4. On $\mathbb{R}^3$ (non-compact), $\triangle_V$ has a continuous spectrum $[0,\infty)$; no spectral threshold arises from geometry alone. Compactness of S^3 is the operative geometric ingredient.

III. THE PRINCIPLE OF SPECTRAL SEPARATION

Principle 1 (Spectral Separation)**.** In a compact gauge theory with a hierarchically organised mode spectrum:

(1) The spectral threshold is $\Delta = \inf\{\omega > 0 : \omega \in \mathrm{spec}(H)\}$, determined by the *lowest* eigenfrequency.

(2) A spectral hierarchy organises excitations *above* the threshold and is logically distinct from the threshold itself.

(3) A sequence $a_n \searrow 0$ describes sub-modes of vanishing energy; its limit is zero and cannot serve as a spectral threshold.

Remark III.1 (Correct role of the Fibonacci sequence). By Principle 1, the objects are:

$$\text{Threshold: } \Delta = \omega_1 = \sqrt{3}/R \quad (\text{geometry of } S^3),$$

$$\text{Hierarchy: } \omega_{F_n} \nearrow \infty \quad (\text{Fibonacci band edges } \Omega_n = \omega_{F_n}).$$

The Fibonacci stratification organises the tower above the threshold (Section IV); it does not generate the threshold.

IV. FIBONACCI SPECTRAL STRATIFICATION

A. Spectral Bands

Definition IV.1 (Fibonacci YM bands)**.** The n-th Fibonacci YM spectral band is

$$\mathcal{B}_n = \{\ell \in \mathbb{Z}_{\geq 1} : F_n \leq \ell < F_{n+1}\}, \quad n \geq 1, \tag{5}$$

with band-edge frequency $\Omega_n = \omega_{F_n} = \sqrt{F_n(F_n+2)}/R$.

Theorem IV.2 (Complete partition)**.** *$\{\mathcal{B}_n\}_{n\geq 1}$ is a partition of $\mathbb{Z}_{\geq 1}$; the corresponding mode subspaces give a complete orthogonal decomposition of the one-particle Hilbert space.*

Proof. The Fibonacci sequence is strictly increasing with $F_n \to \infty$, so $[F_n, F_{n+1})$ are disjoint and cover $\mathbb{Z}_{\geq 1}$. Orthogonality follows from that of vector harmonics at distinct ℓ. □

B. Golden-Ratio Asymptotics

Theorem IV.3 (Golden-ratio scaling)**.** *As $n \to \infty$,*

$$\frac{\Omega_{n+1}}{\Omega_n} = \varphi\big(1 + O(\varphi^{-2n})\big) \to \varphi. \tag{6}$$

Proof. Binet's formula gives $F_n = \varphi^n/\sqrt{5}\cdot(1+O(\varphi^{-2n}))$. Therefore $\Omega_{n+1}^2/\Omega_n^2 = F_{n+1}(F_{n+1}+2)/[F_n(F_n+2)] \sim (F_{n+1}/F_n)^2 \to \varphi^2$. Taking square roots yields (6). □

Remark IV.4. The golden ratio φ is the Perron–Frobenius eigenvalue of the companion matrix $\left(\begin{smallmatrix}1 & 1\\ 1 & 0\end{smallmatrix}\right)$ of the Fibonacci recurrence, applied to the mode labels of the S^3 Laplacian.

C. Natural Minimal-Redundancy Status of the Fibonacci Partition

Theorem IV.5 (Natural minimal-redundancy partition)**.** *Among all non-overlapping gapless partitions of $\mathbb{Z}_{\geq 1}$ into consecutive intervals with convergent ratio $r > 1$, satisfying additive closure ($a_m + a_n = a_k$ for some k) and minimal redundancy, the Fibonacci partition is the unique solution with $r = \varphi$.*

Proof. Additive closure (i.e., that mode-label sums remain within the partition system) with $a_1 = 1$ forces the Fibonacci recurrence $a_{n+2} = a_{n+1} + a_n$ by the Zeckendorf uniqueness theorem [15]. The limit $r = \varphi$ is then the unique Perron–Frobenius eigenvalue of the companion matrix. □

The Fibonacci stratification is therefore the *natural minimal-redundancy* spectral coarse-graining of the S^3 mode spectrum: any other non-overlapping gapless partition with the same self-similar property either violates additive closure or is coarser.

V. THE GEOMETRIC INFRARED SCALE

A. Main Theorem

Theorem V.1 (Geometric infrared scale on S^3)**.** *The spectral threshold of the linearised $SU(N_c)$ Yang–Mills Hamiltonian on $S^3(R)\times\mathbb{R}$ in temporal gauge is*

$$\boxed{\Delta = \inf\{\omega > 0 : \omega \in \mathrm{spec}(H)\} = \frac{\sqrt{3}}{R} > 0.} \tag{7}$$

It is achieved at $\ell = 1$ (lowest mode), is a minimum of the spectrum, and equals $\sqrt{\lambda_1^V(S^3(R))}$ as given by Lemma II.1 and Corollary II.2.

Proof. From (4): $\omega_1^2 R^2 = 3$, so $\omega_1 = \sqrt{3}/R$. For $\ell \geq 2$: $\ell(\ell+2) \geq 8 > 3$, so $\omega_\ell > \omega_1$. All multi-particle energies are sums of $\omega_\ell \geq \omega_1$, giving $\Delta = \omega_1$. □

B. Scope: Geometric Infrared Scale vs. Full Non-Perturbative Gap

Remark V.2 (Finite-volume vs. infinite-volume). Theorem V.1 establishes a geometric infrared scale for the *linearised* theory on the *finite-volume* background $S^3(R)$. The full non-perturbative Yang–Mills mass gap on $\mathbb{R}^3\times\mathbb{R}$ requires controlling the limit $R \to \infty$ in the interacting theory, which lies beyond the scope of this paper.

The interacting theory on S^3 is studied in finite-volume YM perturbation theory [14]; the leading correction to the linearised threshold is positive (Proposition V.3 below).

Proposition V.3 (Positive interacting correction)**.** *In the weak-coupling expansion on $S^3(R)$, the leading non-Abelian correction to the spectral threshold from the quartic $[A,A]^2$ vertex is*

$$\delta\Delta = \frac{c_G g^2(1/R)}{R^3} + O\left(\frac{g^4}{R^5}\right), \quad c_G = \frac{N_c C_2(G)}{8\pi^2}, \tag{8}$$

where $C_2(SU(N_c)) = N_c$ and $g(1/R)$ is the running coupling at scale $1/R$. The coefficient $c_G > 0$, suggesting a positive shift; the full magnitude requires a non-perturbative treatment beyond the weak-coupling expansion.

Proof (within weak-coupling expansion on S^3). The tadpole contribution from the quartic vertex, evaluated via heat-kernel regularisation on S^3 (Appendix C and [10]), gives a self-energy shift $\Pi^{(1)} \propto g^2 C_2(G)/R^3$ for the $\ell = 1$ mode. The sign follows from $C_2(G) > 0$ and the positive semi-definiteness of the one-loop self-energy matrix, within the weak-coupling regime. The expansion is valid for $g^2(1/R) \ll 1$, i.e. in the perturbative regime above the confinement scale. □

Conjecture V.1 (Infinite-volume limit)**.** A positive spectral threshold persists in the limit $R \to \infty$ in the full non-perturbative $SU(N_c)$ Yang–Mills theory on $\mathbb{R}^3\times\mathbb{R}$.

The resolution of Conjecture V.1 for $R \to \infty$ would provide a step toward the Clay Millennium Problem (Section IX).

C. Physical Scale

Corollary V.4 (Physical value)**.** *Setting $R = 1/\Lambda_{\rm QCD}$ (confinement condition, see Section VII) and $\Lambda_{\rm QCD} \approx 300\,\text{MeV}$:*

$$\Delta = \sqrt{3}\,\Lambda_{\rm QCD} \approx 520\,\text{MeV}. \tag{9}$$

This coincides with the effective gluon mass extracted from Dyson–Schwinger analyses [4, 5] ($m_g^{\rm eff} \approx 500\,\text{MeV}$).

VI. THE GLUEBALL SPECTRUM

A. Two-Gluon States at Fibonacci Band Edges

In the leading-order bound-state approximation (two gluons at Fibonacci band edge Ω_n, neglecting binding energy corrections),

$$m_n \approx 2\Omega_n = \frac{2\sqrt{F_n(F_n+2)}}{R}. \tag{10}$$

Binding energy corrections and loop effects are expected at the $O(20\%)$ level (Conjecture V.1).

Theorem VI.1 (Glueball mass hierarchy)**.** *Under approximation* (10)*:*

(a) *Lightest glueball:* $m_1 = 2\sqrt{3}/R = 2\sqrt{3}\,\Lambda_{\rm QCD}$.

(b) $m_{n+1}/m_n \to \varphi$ *as* $n \to \infty$.

(c) *First non-degenerate ratio ($\Omega_1 = \Omega_2$ since $F_1 = F_2 = 1$):*

$$\frac{m_3}{m_1} = \sqrt{\frac{F_3(F_3+2)}{F_1(F_1+2)}} = \sqrt{\frac{7}{3}} \approx 1.528.$$

Proof. (a) and (c): direct computation from (10). (b): $m_n = 2\Omega_n$ and Theorem IV.3. □

B. Quantum Number Assignments

Vector harmonics at level ℓ carry $J = \ell$ under $SO(3) \subset SO(4)$. The dominant J^{PC} assignments for symmetric two-gluon states:

Band n	F_n	Dominant J^{PC}
1, 2	1	0^{++}, 2^{++}
3	2	0^{-+}, 2^{-+}
4	3	1^{+-}, 3^{+-}
5	5	higher-spin

C. Comparison with Quenched Lattice QCD

TABLE I. Glueball masses from quenched lattice QCD [3] and the free-mode S^3 model. Two scale choices: $\Lambda_{\rm QCD} = 300\,\text{MeV}$ ($\overline{\rm MS}$) and $r_0^{-1} = 410\,\text{MeV}$ (Sommer scale). All masses in MeV.

J^{PC}	Lattice	n (F_n)	S^3 (300)	S^3 (410)
0^{++}	1730 ± 80	1 (1)	1039	1421
2^{++}	2400 ± 120	3 (2)	1587	2170
0^{-+}	2590 ± 130	3 (2)	1587	2170
1^{+-}	2940 ± 30	4 (3)	2244	3069
3^{+-}	3550 ± 50	5 (5)	3496	4779

The scale-free ratio $m_3/m_1 = \sqrt{7/3} \approx 1.528$ compares with the lattice value $2400/1730 \approx 1.387$, a $\sim 10\%$ discrepancy attributable to two-body binding energy, loop corrections, and lattice-spacing artefacts neglected in the free-mode approximation.

VII. RENORMALIZATION GROUP AND THE CONFINEMENT RADIUS

The one-loop β-function for $SU(N_c)$ with N_f massless quarks is $\mu\partial g/\partial\mu = -b_0 g^3/(16\pi^2) + O(g^5)$, $b_0 = (11N_c - 2N_f)/3 > 0$. Dimensional transmutation generates $\Lambda_{\rm QCD} = \mu \exp[-8\pi^2/(b_0 g^2(\mu)) + O(g^2)]$.

The identification $R = 1/\Lambda_{\rm QCD}$ is motivated by dimensional transmutation scale matching: $R_{\rm conf}$ is defined as the spatial scale at which the running coupling reaches $O(1)$,

$$g^2(1/\Lambda_{\rm QCD}) \sim O(1), \quad R_{\rm conf} = 1/\Lambda_{\rm QCD} \approx 0.66\,\text{fm}. \tag{11}$$

With this identification, Corollary V.4 gives $\Delta \approx 520\,\text{MeV}$.

VIII. PREDICTIONS

A. Golden-Ratio Glueball Mass Ratio

The primary falsifiable prediction is the asymptotic ratio of consecutive glueball masses at large spin $J = F_n$:

$$\lim_{n\to\infty} \frac{m_{n+1}^{\rm glueball}}{m_n^{\rm glueball}} = \varphi = 1.6180\ldots \tag{12}$$

Current lattice data (limited to $J \le 3$) show ratios between 1.38 and 1.70; the convergence to φ is a prediction testable by future lattice computations at $J \ge 5$.

B. Hagedorn Temperature

The Fibonacci band degeneracies grow as $\mathcal{N}_n \sim \varphi^{3n}$ (Appendix D). This exponential growth implies a Hage-

dorn temperature:

$$T_H = \frac{\Lambda_{\rm QCD}}{\ln\varphi} \approx 2.08\,\Lambda_{\rm QCD} \approx 620\,{\rm MeV} \quad (\Lambda_{\rm QCD} = 300\,{\rm MeV}). \tag{13}$$

C. Summary

IX. DISCUSSION

A. Relation to the Clay Millennium Problem

The Clay problem [1] requires a positive mass gap in infinite-volume $\mathbb{R}^3 \times \mathbb{R}$. Theorem V.1 gives a positive spectral threshold in finite volume $S^3(R) \times \mathbb{R}$.

On $\mathbb{R}^3$, the vector Laplacian has continuous spectrum $[0, \infty)$; any gap requires purely non-perturbative dynamics. On $S^3(R)$, the geometric threshold $\sqrt{3}/R$ already provides a lower bound in the linearised theory, raised further by non-Abelian interactions (Proposition V.3). The compact geometry separates the problem into two components: a geometric contribution (proved here) and a dynamical one (Conjecture V.1).

Open problems:

(1) Does a positive mass gap persist as $R \to \infty$?
(2) Can the two-gluon approximation (10) be systematically improved via the operator algebra on S^3?
(3) Is $T_H = \Lambda_{\rm QCD}/\ln\varphi$ identifiable as a deconfinement scale?

B. Comparison with Existing Approaches

Finite-volume YM on T^3 [14]: compact geometry gives discrete spectrum, but $\pi_1(T^3) = \mathbb{Z}^3$ violates (M2) and $SO(3)$ isometry is less symmetric than $SO(4)$ on S^3.

Lattice QCD [2, 3]: numerical evidence for the gap; no geometric mechanism identified.

Dyson–Schwinger equations [5]: dynamical gluon mass $\sim 500\,{\rm MeV}$, consistent with $\Delta \approx 520\,{\rm MeV}$.

Bag model [16]: confinement with a phenomenological bag radius; in the present framework, $R = 1/\Lambda_{\rm QCD}$ is fixed by the running coupling.

AdS/CFT: holographic gap from IR bulk geometry; Fibonacci spectral structure does not arise naturally.

X. CONCLUSIONS

This paper establishes a geometric infrared scale for $SU(N_c)$ Yang–Mills theory in finite spatial volume $S^3(R) \times \mathbb{R}$:

1. *Lemma II.1* (general compact 3-manifold): the linearised YM Hamiltonian has purely discrete spectrum with positive threshold $\Delta(M) = \sqrt{\lambda_1^V(M)}$.
2. *Theorem V.1* (S^3 specialisation): $\Delta = \sqrt{3}/R = \sqrt{3}\,\Lambda_{\rm QCD} \approx 520\,{\rm MeV}$, consistent with Dyson–Schwinger gluon mass estimates.
3. *Principle 1* (spectral separation): the infrared scale is determined by the lowest eigenfrequency; a Fibonacci hierarchy organises higher excitations above it.
4. *Theorem IV.5* (natural minimal-redundancy Fibonacci partition): the Fibonacci stratification is a natural minimal-redundancy self-similar partition of the mode spectrum.
5. *Theorem VI.1* (glueball spectrum): the mass ratio $m_3/m_1 = \sqrt{7/3} \approx 1.528$ is consistent with quenched lattice QCD within 10%; $m_{n+1}/m_n \to \varphi$ is a prediction testable at higher spin.

The formula $\Delta = \sqrt{3}\,\Lambda_{\rm QCD}$ gives the *geometric infrared scale* of Yang–Mills theory on the minimal compact 3-manifold. Whether it survives as an infinite-volume non-perturbative mass gap is Conjecture V.1, and remains open.

DATA AVAILABILITY

This manuscript reports a theoretical analysis. No new experimental or observational data sets were generated or analyzed. All derivations and numerical estimates needed to assess the claims of the paper are contained in the manuscript itself.

[1] A. Jaffe and E. Witten, *Yang–Mills Existence and Mass Gap*, Clay Math. Inst. Millennium Prize Problems (2000).
[2] C. J. Morningstar and M. J. Peardon, Glueball spectrum from an anisotropic lattice study, Phys. Rev. D **60**, 034509 (1999).
[3] Y. Chen et al., Glueball spectrum and matrix elements on anisotropic lattices, Phys. Rev. D **73**, 014516 (2006).
[4] J. M. Cornwall, Dynamical mass generation in continuum QCD, Phys. Rev. D **26**, 1453 (1982).
[5] A. C. Aguilar et al., Schwinger displacement of the quark–gluon vertex, Eur. Phys. J. C **80**, 154 (2020).
[6] G. Perelman, The Entropy Formula for the Ricci Flow and its Geometric Applications, arXiv:math/0211159 (2002).
[7] G. Perelman, Ricci Flow with Surgery on Three-Manifolds, arXiv:math/0303109 (2003).

TABLE II. Summary of the main observables, their leading predictions in the compact-S^3 framework, and present status.

Observable	Prediction	Status
Infrared scale Δ	$\sqrt{3}\,\Lambda_{\rm QCD} \approx 520\,\text{MeV}$	Consistent with DSE
$m(0^{++})$	$2\sqrt{3}\,\Lambda_{\rm QCD} \approx 1.4\,\text{GeV}$	Consistent with lattice
m_3/m_1	$\sqrt{7/3} \approx 1.528$	Lattice 1.39 ($\sim 10\%$)
m_{n+1}/m_n as $n \to \infty$	$\varphi = 1.618$	Future lattice test
Hagedorn temperature	$\Lambda_{\rm QCD}/\ln\varphi \approx 620\,\text{MeV}$	Testable

[8] G. Perelman, Finite Extinction Time for the Solutions to the Ricci Flow on Certain Three-Manifolds, arXiv:math/0307245 (2003).

[9] R. Camporesi and A. Higuchi, Spectral functions and zeta functions in hyperbolic spaces, J. Math. Phys. **35**, 4217 (1994).

[10] J. S. Dowker and R. Critchley, Effective Lagrangian and energy–momentum tensor in de Sitter space, Phys. Rev. D **13**, 3224 (1976).

[11] M. E. Taylor, *Partial Differential Equations, Vol. 2*, Springer (1996).

[12] P. B. Gilkey, *Invariance Theory, the Heat Equation, and the Atiyah–Singer Index Theorem*, 2nd ed., CRC Press (1995).

[13] M. Berger, P. Gauduchon, and E. Mazet, *Le spectre d'une variété riemannienne*, Springer Lecture Notes Math. **194** (1971).

[14] M. Lüscher, Some analytic results concerning the mass spectrum of Yang–Mills gauge theories on a torus, Nucl. Phys. B **219**, 233 (1983).

[15] E. Zeckendorf, Représentation des nombres naturels par une somme de nombres de Fibonacci, Bull. Soc. R. Sci. Liège **41**, 179 (1972).

[16] A. Chodos, R. L. Jaffe, K. Johnson, C. B. Thorn, and V. F. Weisskopf, New extended model of hadrons, Phys. Rev. D **9**, 3471 (1974).

[17] R. L. Workman et al. (Particle Data Group), Review of Particle Physics, PTEP **2022**, 083C01 (2022).

[18] A. Preece and B. Batenin, Fractal Nature of Gluons and the Mass Gap in Yang–Mills Quantum Theory, Zenodo (2025), doi:10.5281/zenodo.14905887.

[19] R. Hagedorn, Statistical thermodynamics of strong interactions at high energies, Nuovo Cimento Suppl. **3**, 147 (1965).

[20] CMB-S4 Collaboration, CMB-S4 Science Book, arXiv:1610.02743 (2016).

[21] LiteBIRD Collaboration, Probing Cosmic Inflation with the LiteBIRD CMB Polarization Survey, PTEP **2023**, 042F01 (2023).

Appendix A: Vector Hyperspherical Harmonics on S^3

We parametrise $S^3(R)$ by $(\chi, \vartheta, \phi) \in [0,\pi] \times [0,\pi] \times [0, 2\pi)$: $ds^2 = R^2[d\chi^2 + \sin^2\chi(d\vartheta^2 + \sin^2\vartheta\, d\phi^2)]$. The scalar Laplacian has eigenvalues $\ell(\ell+2)/R^2$, degeneracy $(\ell+1)^2$.

On $S^3 \cong SU(2)$ with isometry group $SO(4) \cong SU(2)_L \times SU(2)_R$, transverse vector harmonics at level ℓ lie in the $(\ell/2, \ell/2)$ representation, with degeneracy $d_\ell = 2(\ell+1)^2$ per colour. The connection (rough) Laplacian in the transverse sector equals the scalar Laplacian on S^3 (see [9]), giving eigenvalues $\lambda_\ell^V = \ell(\ell+2)/R^2$ and minimum $\lambda_1^V = 3/R^2$.

Appendix B: Complete Fibonacci Spectral Table

TABLE III. Fibonacci spectral data. Physical masses $m_n = 2\sqrt{F_n(F_n+2)} \times r_0^{-1}$ with $r_0^{-1} = 410\,\text{MeV}$. Ratio $= \Omega_{n+1}/\Omega_n \to \varphi = 1.6180$.

n	F_n	$\Omega_n R$	m_n (MeV)	Ratio
1	1	1.732	1422	—
2	1	1.732	1422	1.000
3	2	2.828	2320	1.633
4	3	3.873	3177	1.369
5	5	5.916	4852	1.527
6	8	8.944	7334	1.512
7	13	13.96	11450	1.561
8	21	21.98	18027	1.575
9	34	34.47	28267	1.569
10	55	55.95	45877	1.623
11	89	89.99	73793	1.608
12	144	144.9	118818	1.613

Ratios converge to $\varphi = 1.6180$ from alternating sides; agreement to within 0.03% by $n = 12$.

Appendix C: One-Loop Correction on S^3

The tadpole self-energy of the $\ell = 1$ mode from the quartic vertex, computed via heat-kernel regularisation [10]:

$$\Pi^{(1)} = \frac{g^2 C_2(G)}{R^3} \sum_{\ell=1}^{\infty} \frac{2(\ell+1)^2}{\sqrt{\ell(\ell+2)}}\, e^{-s\ell(\ell+2)}. \tag{C1}$$

For small s: $\sum 2(\ell+1)^2 e^{-s\ell^2} \sim \sqrt{\pi}/(2s^{3/2})$. After zeta-function regularisation (finite renormalised constant $c_{\rm reg} > 0$), the shift $\delta\omega_1 = c_G g^2/R^3 > 0$ confirms Proposition V.3.

Appendix D: Hagedorn Temperature

The mode count in band $\mathcal{B}_n$ is $\mathcal{N}_n \approx (2/3)(N_c^2-1)(F_{n+1}^3 - F_n^3) \sim \varphi^{3n}$. The partition function $Z(\beta) = \prod_n (1-e^{-\beta\Omega_n})^{-\mathcal{N}_n}$ has a Hagedorn singularity [19] when $\mathcal{N}_n \sim e^{\beta\Omega_n}$, i.e. when $3\ln\varphi = \beta/(\sqrt{5}\,R)$ per Fibonacci step:

$$T_H = \frac{\Lambda_{\rm QCD}}{\ln\varphi} = \frac{\Lambda_{\rm QCD}}{\ln[(1+\sqrt{5})/2]} \approx 2.08\,\Lambda_{\rm QCD}. \tag{D1}$$

Appendix E: Wilson Loop and Area Law

In the strong-coupling expansion on S^3, the Wilson loop satisfies

$$\langle W[\mathcal{C}]\rangle \sim e^{-\sigma L T_0}, \quad \sigma = \frac{\ln g^2}{R^2} + O(g^{-2}). \tag{E1}$$

The lowest Fibonacci trajectory ($F_1 = 1$) gives string tension $\sigma_1 = \Lambda_{\rm QCD}^2 \approx 0.09\,\mathrm{GeV}^2$, within a factor of 2 of the phenomenological value $\sigma_{\rm phys} \approx 0.18\,\mathrm{GeV}^2$.